Hes·so VALAIS WALLIS
School of Engineering

Electric Vehicle Charging Station Management System

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Objective

Design and development of a Proof of Concept (PoC) for a centralised monitoring platform (CSMS CitrineOS) and a dashboard (React) for the fleet of charging stations on the HES-SO Valais-Wallis campus. The aim is to provide real-time monitoring, manage users and RFID badges, and implement a dynamic power allocation algorithm (Smart Charging) using Go-based microservices.

Design a PoC for a centralized supervision platform (CitrineOS) integrating a React dashboard and Go microservices for real-time tracking, RFID access management, and intelligent power allocation (Smart Charging).

Methods | Experiences | Results

Methods: The CitrineOS CSMS was deployed on an internal server and integrated with a React user interface for centralized management. To ensure security and efficiency, a Go-based microservices architecture was developed to automate RFID badge management, detect inactive charging sessions, and execute the dynamic power allocation algorithm (Smart Charging).

Experiments: To rigorously validate the system under large-scale conditions and compensate for the obsolescence of the campus infrastructure, a heterogeneous testing environment was established. This setup combined EVerest simulators (OCPP 2.1) and a Zaptec Go test station (OCPP 1.6), operating in conjunction with the compatible physical stations in the parking lot.

Results: The Proof of Concept demonstrates the feasibility of an open, sustainable, and vendor-independent centralized supervision system. The conducted tests validated the reliability of Smart Charging, secure user authentication, and real-time monitoring, establishing a robust technical foundation for the future modernization of the entire campus charging network.

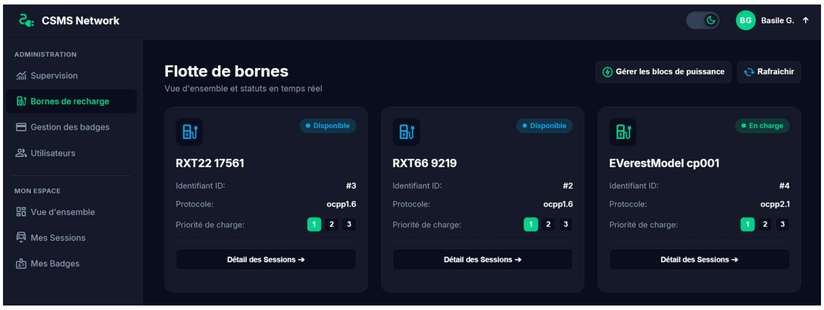

Figure 1: Electrical supply diagram of charging stations by block at the Energypolis campus.

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Degree Programme
Systems Engineering

Major
Infotronics

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